

Name: Dog Paw Washer _____ Date: 3/9/2026 _____
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MyDesign Portfolio

Element A Initial Template

Element A: Presentation and Justification of the Problem

Problem Statement and Details (A1, A2, A5)

Dogs are common household companions in the United States, with approximately 65 million U.S. households owning at least one dog, meaning issues related to pet cleanliness affect a large portion of the population (American Pet Products Association, 2023). One everyday challenge for dog owners occurs when dogs return from outdoor walks, yards, or parks with dirty paws that carry mud, bacteria, allergens, and environmental contaminants into the home. Research shows that dogs' paws can collect and transport microorganisms, soil particles, and pollutants from outdoor surfaces such as sidewalks, parks, and roads into indoor environments, where these contaminants may spread onto floors, carpets, and furniture (Lefebvre et al., 2017). This problem occurs most frequently after outdoor activity in wet, muddy, or winter conditions when paws accumulate water, dirt, and road salt. In colder regions, road salt and de-icing chemicals used on sidewalks and streets can irritate dog's paws and can also be tracked inside homes, contributing to damage on flooring and potential health concerns for pets (American Veterinary Medical Association, 2022). Because dogs often enter homes immediately after walks, owners must frequently clean floors, wipe paws manually, or bathe their dogs, adding time and effort to daily routines. Surveys of pet owners indicate that maintaining cleanliness and preventing dirt from being tracked inside is a common concern, particularly for households with active dogs that spend significant time outdoors (American Kennel Club, 2021). As dog ownership continues to remain widespread, the issue of outdoor contaminants being carried indoors by dog's paws occurs regularly in households across communities, making it an ongoing inconvenience for pet owners and highlighting the need for efficient solutions to help keep homes clean.

Stakeholder Concerns and Connections (A3, A4)

Primary Stakeholders include but are not limited to:

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- Dog Owners
- Dogs (or other pets)
- Family members living in the same home
- Cleaning/Maintenance workers or homeowners responsible for cleaning.
- Veterinarians and pet health professionals.

Concerns Of:

Dog Owners - Dog owners are the most directly affected because they must deal with the time and effort required to clean floors, carpets, and furniture after their dogs come inside with dirty paws. They may also worry about the cost of maintaining a clean home and preventing damage to flooring.

Dogs (or other pets) - Dogs themselves can be affected because dirt, chemicals, and road salt stuck on their paws can cause irritation, dryness, or injury. According to the American Veterinary Medical Association, winter road salts and other outdoor chemicals can irritate dogs' paws and cause discomfort.

Family Members in the Home - Other people living in the home may be affected by dirt, bacteria, or allergens brought inside by dogs' paws. This can create hygiene concerns, especially for children, elderly people, or individuals with allergies or immune disorders.

Cleaning or Maintenance Workers - People responsible for maintaining the home must spend additional time cleaning floors, carpets, and rugs after dogs track in mud or debris, which increases workload and maintenance effort.

Veterinarians and Pet Health Professionals - Veterinarians may see dogs with irritated paws or skin problems caused by exposure to environmental contaminants such as road salt or chemicals picked up while walking outside.

These concerns show that the problem affects both home cleanliness and pet health, making it important to develop a solution that helps reduce the amount of dirt and contaminants dogs bring indoors.

Problem Importance (A6)

This problem is important to dog owners who are stakeholders because it keeps their house and furniture clean while living with a dog that can go on healthy walks in healthy



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environments outside. Currently 65.1 million households, almost 50%, in the United States own dogs, meaning that this problem is widespread, and impacts the lives of many. Perhaps more importantly, the U.S. dog population is about 89 million dogs, meaning that the sanity and safety of millions of dogs can benefit from a solution to this problem (APPA, 2023). Further, 90% of dogs carry bacteria on their paws or fur, and can transport soil, pesticides, heavy metals, environmental pollutants, road salt, and de-icing chemicals into homes, putting the cleanliness and health of dog owners at risk as well (Lefebvre et al., 2017)(American Veterinary Medical Association, 2022). This problem impacts millions, and affects the health and well being of dog owners and dogs themselves. A solution to this problem could solve health complications associated with pollutants in the home, and increase dogs' quality of life.

Reference List:

1. American Pet Products Association. (2023). *APPA National Pet Owners Survey*.
<https://www.americanpetproducts.org>
(Professional organization)
2. Lefebvre, S., et al. (2017). "Pets as sources of household contamination and infection risk." *Veterinary Clinics of North America: Small Animal Practice*.
(Peer-reviewed journal)
3. American Veterinary Medical Association. (2022). *Winter Pet Safety and Paw Protection*.
<https://www.avma.org>
(Professional veterinary organization)
4. American Kennel Club. (2021). *Keeping Your Home Clean With Dogs*.
<https://www.akc.org>
(Major national organization focused on dog care)